SIMATS Online Programming Lab

PYTHON PROGRAMMING

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Q1 **Majority Element -List** 10 marks

Write a Python function to check whether a list contains a majority element.

INPUT:

2 2 1 2 3 2 2

OUTPUT:

2

Your Code

```python
def majority_element(arr):
    count = {}
    n = len(arr)

    for num in arr:
        count[num] = count.get(num, 0) + 1
        if count[num] > n // 2:
            return num

    return None

# Reading input
nums = list(map(int, input().split()))
result = majority_element(nums)
print(result)
```

Input

```
2 2 1 2 3 2 2
```

Output

```
2
```

Visible Test Cases

Test Case 1

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Write a Python program to find a peak element in a list of integers using list. [A peak element is an element that is greater than or equal to its adjacent elements.]

INPUT:

1 3 20 4 1 0

OUTPUT:

20

Your Code

```
nums = list(map(int, input().split()))
n = len(nums)
peak = nums[0]
for i in range(n):
    if i == 0:
        if n == 1 or nums[i] >= nums[i + 1]:
            peak = nums[i]
            break
    elif i == n - 1:
        if nums[i] >= nums[i - 1]:
            peak = nums[i]
            break
    else:
        if nums[i] >= nums[i - 1] and nums[i] >= nums[i + 1]:
            peak = nums[i]
            break
print(peak)
```

Input

```
1 3 20 4 1 0
```

Output

```
20
```

Visible Test Cases

Test Case 1 ✔

10.00 / 10 marks

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Write a recursive Python function to determine whether a given number is a Lucky Number.

A Lucky Number is a number whose sum of digits is reduced repeatedly until a single digit is obtained, and the final digit is 7.

INPUT1:

34

OUTPUT1:

Lucky Number

INPUT2:

123

OUTPUT2:

Not a Lucky Number

Your Code

```python
def is_lucky_number(n):
    def recursive_sum(x):
        if x < 10:
            return x
        s = 0
        while x > 0:
            s += x % 10
            x //= 10
        return recursive_sum(s)
    return recursive_sum(n) == 7
# Read input
n = int(input().strip())
print("Lucky Number" if is_lucky_number(n) else "Not a Lucky Number")
```

Input

```
34
123
```

Output

```
Lucky Number
```

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Q6 Second Largest and Second Smallest - Tuple

10 marks

Write a Python program to find the second largest and second smallest elements in a list using a tuple.

INPUT:

10 20 5 40 15 30

OUTPUT:

10

30

Your Code

```python
arr = list(map(int, input().split()))
# second smallest
sorted_arr = sorted(arr)
second_smallest = sorted_arr[1]
# second largest
second_largest = sorted_arr[-2]
result = (second_smallest, second_largest)
print(result[0])
print(result[1])
```

Input

```
10 20 5 40 15 30
```

Output

```
10
30
```

Q7 Maximum Element - Tuple

10 marks

Write a Python program to find the largest element in a tuple.

INPUT:

12 45 8 67 23

OUTPUT:

67

Your Code

```python
t = tuple(map(int, input().split()))
print(max(t))
```

Input

```
12 45 8 67 23
```

Output

```
67
```

Q10 **Sum & Average - Tuple** 10 marks

Write a Python program to find the sum and average of the elements in a tuple.

INPUT:

10 20 30 40 50

OUTPUT:

150

30.0

Your Code

```
t = tuple(map(int, input().split()))
s = sum(t)
avg = s / len(t)
print(s)
print(avg)
```

Input

```
10 20 30 40 50
```

Output

```
150
30.0
```

Q8 Frequency of Each Element - List

10 marks

Write a Python program to find the frequency of each element in an array using List.

INPUT:

1 2 3 2 4 1 5 2

OUTPUT:

1 = 2

2 = 3

3 = 1

4 = 1

5 = 1

Your Code

```
arr = list(map(int, input().split()))
freq = {}
for x in arr:
    freq[x] = freq.get(x, 0) + 1
for key in sorted(freq.keys()):
    print(f"{key} = {freq[key]}")
```

Input

```
1 2 3 2 4 1 5 2
```

Output

```
1 = 2
2 = 3
3 = 1
4 = 1
5 = 1
```

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Q9 **Remove Duplicate Elements - Tuple** 10 marks

Write a Python program to remove duplicate elements from a tuple.

INPUT:

10 20 30 20 40 10 50

OUTPUT:

(40, 10, 50, 20, 30)

Your Code

```
t = tuple(map(int, input().split()))
result = tuple(dict.fromkeys(t))  # removes duplicates while preserving
print(result)
```

Input

```
10 20 30 20 40 10 50
```

Output

```
(10, 20, 30, 40, 50)
```